

Majorana + Berezinskii Kosterlitz Thouless using Quantum Monte Carlo

Scientific question:

- Find the classical phase diagram of 2D plasma with charges $\zeta, \zeta' = \pm 1$ and interaction

$$V_{\zeta, \zeta'}(\mathbf{r}, \mathbf{r}') = -\zeta\zeta'K \ln(|\mathbf{r} - \mathbf{r}'|/\xi) - T \ln(2 \cosh(E_0/T e^{-|\mathbf{r} - \mathbf{r}'|/\xi_F})) \stackrel{|\mathbf{r} - \mathbf{r}'| \rightarrow \xi^+}{\simeq} \left[-\zeta\zeta' \frac{K}{\xi} + \frac{E_0}{\xi_F} \right] |\mathbf{r} - \mathbf{r}'| - E_0 \quad (1)$$

(For the free energy of a two-vortex system, note that the entropic contribution leads to $-\zeta\zeta'K \rightarrow -\zeta\zeta'K - T$. Note that the term E_0/ξ can reverse the sign of the small $\mathbf{r} - \mathbf{r}'$ interaction from repulsive to attractive.)

- Parameter regime of relevance
 - K/T - arbitrary (transition close to K/T expected).
 - $E_0/T \gg 1$.
 - $\xi_F/\xi \gg 1$.
- Expected phases:
 - $K/T \gg 1$: charges are tightly bound in pairs
 - $K/T \ll 1$: charges unbound (i.e. plasma state)

For $E_0/T = 0$ phase transition of BKT type.

- Question:
 - Does the nature of the phase transition change as E_0/T is cranked up?
 - Are there additional phases appearing (e.g. phases in which vortex pairs are locked together but proliferate)

Steps to proceed for a student:

1. Learn Metropolis Algorithm for Monte Carlo and phase transitions.
 - Familiarize yourself with the 2D Ising model
 - Learn how to play solve it with Metropolis algorithm (see e.g. tutorials [1 (Excercise Sheet 1 if you can open it), 2, 3])
2. Learn basics about theory of phase transitions
 - 1st vs. second order (use, e.g. review by Kurt Binder)
 - BKT transission (see eg. correpsonding chapter in textbook by A Altland and B Simons – the renormalization group might be a bit advanced, but there is a qualitative argument in the beginning of the chapter)
3. Learn BKT physics
 - learn basics about Monte Carlo approach to Coulomb Gas problem in various dimensions
 - Find out [thesis](#)
4. Monte Carlo simulate the two-component 2D plasma with the interaction given in Eq. (1).